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Physics-Informed Neural Networks with Hard Constraints for Axial Temperature Distribution Estimation of Lithium-Ion Batteries

Lingqing Guo [1], Kangliang Zheng [1], Xiucheng Wu [1], Jinhong Wang [1], Xiaofeng Lai [1], Peiyuan Deng [1], Lv He [2], Yuan Cao [2], Chengying Zeng [3] and Xiaoyu Dai [3,*]

[1] Huizhou Power Supply Bureau, Guangdong Power Grid Corporation, Huizhou 516000, China; glqssg@163.com (L.G.); 18819460662@139.com (K.Z.); 18927336830@163.com (X.W.); 18948231376@189.cn (J.W.); xiaofeng.lai3999@foxmail.com (X.L.); 13809834391@139.com (P.D.)
[2] School of Automation, Central South University, Changsha 410083, China; 254612137@csu.edu.cn (L.H.); caoyuan3116@csu.edu.cn (Y.C.)
[3] School of Electrical Engineering and New Energy, Three Gorges University, Yichang 443002, China; 202310090127@ctgu.edu.cn
* Correspondence: 202310730154@ctgu.edu.cn; Tel.: +86-19071390724

Abstract

Accurate estimation of the internal spatial-temporal temperature distribution is crucial for the safety and performance management of lithium-ion batteries. However, traditional lumped parameter models overlook spatial gradients, while numerical methods for partial differential equations (PDEs) incur high computational costs. This paper proposes a hard constraint physics-informed neural network (HCPINN) framework for the real-time reconstruction of the axial temperature field in 18,650 cylindrical batteries. By restructuring the neural network's solution space through distance functions, the Robin boundary conditions are strictly embedded as hard constraints, ensuring exact satisfaction of the prescribed Robin boundary conditions within the mathematical model and eliminating boundary loss terms. An electro-thermal coupled model considering the Arrhenius effect and state-of-charge (SOC) dependent internal resistance is integrated into the loss function to capture the nonlinear heat generation dynamics. Experimental validation across discharge rates from 1C to 4C demonstrates that the HCPINN achieves high estimation accuracy with a mean absolute error (MAE) below 0.34 °C. Furthermore, by leveraging the continuous differentiability of the model, this study quantifies the evolution of spatial temperature gradients and reveals the ideal heat transfer coefficients required for thermal equilibrium are inverted, providing a quantitative basis for the design of advanced battery thermal management systems (BTMS).

Keywords: lithium-ion batteries; temperature distribution estimation; physics-informed neural networks (PINN); hard constraints; electro-thermal coupling; thermal management systems

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1. Introduction

As the global transition towards sustainable energy accelerates, lithium-ion batteries (LIBs) have become the predominant energy storage solution for electric vehicles (EVs) and smart grids due to their high energy density, low self-discharge rate, and long operational life [1–6]. However, the performance, reliability, and safety of LIBs are highly sensitive to their operating temperature [7]. During aggressive charging or high-rate discharge cycles, complex electrochemical reactions and internal resistance lead to substantial heat generation [8]. If

this heat is not managed effectively, it can result in significant spatial temperature gradients, accelerated capacity fade, and in extreme cases, thermal runaway [9–12].

Specifically, for cylindrical cells like the 18,650, the heat transfer path along the axial direction is often neglected in simplified models, yet it plays a critical role in localized aging and stress distribution [13,14]. Traditional battery management systems (BMS) primarily rely on surface-mounted sensors, which fail to capture the real-time internal temperature field due to the thermal insulation of the casing and the time-lag effect of heat conduction. Therefore, developing a high-fidelity and computationally efficient method to reconstruct the axial temperature distribution is of paramount importance for the next generation of intelligent thermal management systems.

The quest for accurate battery thermal models has led to the development of several distinct methodologies. Traditional lumped parameter models are widely used in commercial BMS due to their minimal computational load [15]. However, by treating the battery as a single thermal point, they fail to account for the internal temperature gradients that are critical during high-rate operations [16]. To resolve spatial details, partial differential equation (PDE) models, typically solved via finite element analysis (FEA) or finite volume methods (FVM), offer high-fidelity spatial-temporal resolution [17–19]. Despite their accuracy, the high computational cost and the requirement for precise physical parameters (such as anisotropic thermal conductivity) limit their deployment in real-time online estimation [20].

In recent years, the emergence of physics-informed neural networks (PINNs) has provided a promising approach [21–23]. By integrating the heat conduction PDE directly into the loss function of a neural network, PINNs combine the flexibility of data-driven methods with the interpretability of physical laws [24]. Nevertheless, most existing PINN frameworks for battery modeling treat boundary conditions as soft constraints through penalty terms in the loss function. This approach often leads to a conflict between the PDE loss and boundary loss during optimization, resulting in boundary leakage or slow convergence [25–27]. For battery systems where surface heat dissipation significantly dictates internal states, these soft-constrained models often struggle to maintain physical consistency at the boundaries [28]. This gap necessitates the development of a more robust architecture that can strictly enforce boundary physics without relying on hyperparameter tuning.

To overcome the inherent limitations of traditional PINNs, specifically the precision loss at boundaries and the tedious manual tuning of penalty weights, this study develops a hard constraint physics-informed neural network (HCPINN) framework. The core innovations of this work are three-fold:

1. Unlike conventional PINN-based battery thermal models that impose boundary conditions through soft penalty losses, this study constructs a distance-function-based solution structure to embed the Robin heat-exchange boundary condition directly into the network output. As a result, the boundary condition is satisfied by construction rather than by loss minimization.
2. The proposed reconstruction maps the unconstrained neural network output onto a physically admissible manifold, where the boundary temperature-gradient relation is locked by the thermal conductivity and convection coefficient. This design removes the boundary loss term and reduces the need for empirical tuning of boundary penalty weights.
3. The hard-constrained architecture is coupled with an SOC- and temperature-dependent heat generation model, allowing continuous axial temperature field reconstruction from sparse surface measurements under 1C–4C discharge conditions.

4. Beyond temperature estimation, the differentiable reconstructed field is used to invert the critical convection coefficient, providing a quantitative indicator of cooling demand for BTMS design.

The remainder of this paper is structured as follows: Section 2 formulates the axial thermal PDE and the electro-thermal coupling mechanism. Section 3 delineates the HCPINN framework, focusing on the mathematical derivation of hard constraints. Section 4 describes the experimental setup and discusses the estimation results, physical consistency, and the inversion of heat transfer coefficients. Finally, Section 5 concludes the study.

2. Mathematical Modeling and Problem Formulation

To accurately characterize the internal thermal dynamics of LIBs, it is essential to establish a high-fidelity physical model that balances computational efficiency with spatial resolution. For cylindrical 18,650 cells, while radial heat conduction is often the primary focus, the axial temperature gradient significantly influences localized aging and overall thermal safety under high-power loads. In this section, a one-dimensional (1D) axial thermal model is formulated based on the principles of heat transfer, treating the battery as a continuous medium with anisotropic thermal properties.

The modeling framework integrates the energy conservation law with a nonlinear heat generation mechanism. By considering the coupling effects of the SOC and the Arrhenius temperature dependence on internal resistance, the proposed model aims to provide a rigorous physical basis for the implementation of HCPINN. This mathematical foundation ensures that the neural network's predictions are strictly governed by the underlying thermodynamic laws of the battery system.

2.1. Axial Heat Conduction Model

For cylindrical 18,650 LIBs, considering its high aspect ratio and the core-to-surface heat transfer path, the temperature distribution along the axial direction (x-axis) is modeled using the 1D heat conduction equation. Assuming the battery material is a continuous medium with an equivalent axial thermal conductivity λ, the spatiotemporal evolution of the temperature $T(x,t)$ is governed by the following PDE:

$$\rho C_p \frac{\partial T(x,t)}{\partial t} = \lambda \frac{\partial^2 T(x,t)}{\partial x^2} + \dot{Q}(t) \quad x \in [0, L],\ t \in [0, t_{end}] \tag{1}$$

where ρ denotes the equivalent density, C_p is the specific heat capacity, L is the axial height of the cell, and $\dot{Q}(t)$ represents the volumetric heat generation rate.

To solve this PDE, the initial and boundary conditions must be specified. The initial temperature is assumed to be uniform and equal to the ambient temperature:

$$T(x, 0) = T_{amb} \tag{2}$$

At the boundaries ($x = 0$ and $x = L$), the heat exchange between the cell and the environment is described by Robin boundary conditions (convective heat transfer):

$$-\lambda \frac{\partial T}{\partial x} = h_x (T_{amb} - T(x,t)) \quad x = 0, L \tag{3}$$

where h_x is the convective heat transfer coefficient and T_{amb} is the ambient temperature. These boundary constraints are critical as they represent the physical interaction with the BTMS and will be treated as hard constraints in the subsequent HCPINN framework.

The axial-only model is adopted in this study because the experimental sensors are distributed along the height direction of the 18,650 cell and the main objective is to recon-

struct the measured axial temperature evolution under a controlled ambient environment. This formulation does not imply that radial gradients are negligible in all cylindrical cells. Under high-rate operation, non-uniform cooling, large-format cells, or module-level conditions, radial gradients may become dominant. Therefore, the present model should be regarded as a simplified axial reconstruction framework for the tested scenario, while coupled radial-axial or 2D/3D thermal modeling will be pursued in future work.

2.2. Nonlinear Heat Generation Mechanism

The volumetric heat generation rate $\dot{Q}(t)$ in (1) is a critical source term that determines the thermal behavior of LIBs. According to the classic Bernardi equation, the heat generated during the discharge process primarily consists of irreversible Joule heat and reversible entropic heat. Considering the high-rate discharge conditions in this study, the irreversible heat dominates, and the heat generation rate can be expressed as:

$$\dot{Q}(t) = \frac{I^2(t)R_{int}(T, SOC) - I(t)T(t)\frac{dU_{ocv}}{dt}}{V_{batt}} \tag{4}$$

where $I(t)$ is the discharge current, V_{batt} is the active volume of the battery, R_{int} is the internal resistance, and $\frac{dU_{ocv}}{dt}$ is the entropy coefficient.

To account for the dynamic changes in internal resistance during discharge, R_{int} is modeled as a nonlinear function of SOC and temperature T. The temperature dependence follows the Arrhenius law, while the SOC dependence is captured through a high-order polynomial fit based on experimental data:

$$R_{int}(T, SOC) = R_{ref}(SOC) \cdot \exp\left[\frac{E_a}{R_g}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right] \tag{5}$$

$$R_{ref}(SOC) = \sum_{i=0}^{n} a_i \cdot SOC^i \tag{6}$$

where R_{ref} is the reference resistance at temperature T_{ref}, E_a is the activation energy, R_g is the universal gas constant, and a_i are the fitting coefficients. By coupling this nonlinear heat source with the axial thermal PDE, the HCPINN can more accurately reflect the electro-thermal coupling characteristics under diverse operating conditions.

To determine the specific parameters in (6), experimental characterization of the cell resistance was performed under various thermal and electrical conditions. The internal resistance data were obtained through pulse-current tests under different SOC and temperature conditions. The cell was discharged to predefined SOC levels and then rested in a thermostatic chamber until thermal equilibrium was reached at 25 °C, 35 °C, and 45 °C. At each SOC-temperature point, a short current pulse was applied, and R_{int} was calculated from the instantaneous voltage drop using $R_{int} = \Delta V / \Delta I$. Each test was repeated three times, and the averaged resistance value was used for fitting. The temperature dependence was described by the Arrhenius relationship, while the SOC dependence was fitted using a high-order polynomial. The fitting coefficients were identified by least-squares regression, achieving $R^2 > 0.99$ within the tested operating range.

Figure 1 illustrates the fitting results of the internal resistance R_{int} across a wide range of SOC (0% to 100%) and temperatures (25 °C to 45 °C). It can be observed that the resistance exhibits a significant nonlinear increase at low SOC levels and a typical Arrhenius temperature-dependent behavior. The high fitting accuracy ($R^2 > 0.99$) ensures that the heat generation source term $\dot{Q}(t)$ in HCPINN framework is physically grounded.

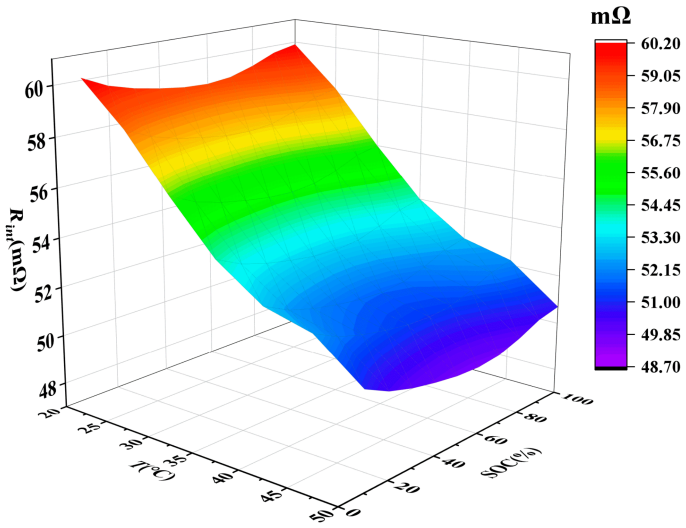


Figure 1. Experimental and fitting results of R_{int} as a function of SOC and temperature.

As shown in Figure 2, the comparison between experimental and fitted resistance values shows that the data points are closely distributed around the 1:1 reference line, indicating that the fitted resistance model can reproduce the measured SOC- and temperature-dependent resistance characteristics. The residuals in Figure 2 remain centered near zero across the investigated SOC range, with no obvious systematic deviation at high, medium, or low SOC states. The corresponding fitting statistics are R^2 = 0.9977, RMSE = 0.42 mΩ, and MAE = 0.32 mΩ, demonstrating that the fitted resistance model provides sufficient accuracy for constructing the heat-generation source term in the subsequent HCPINN formulation.

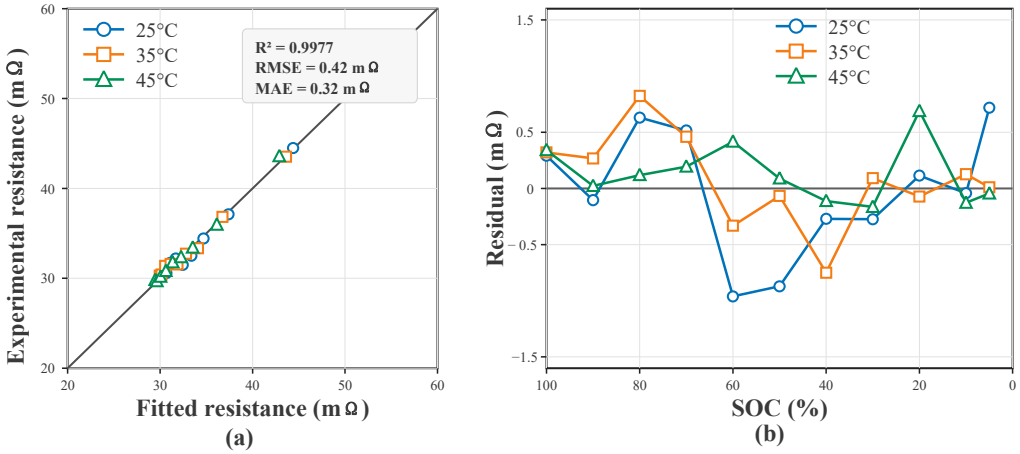


Figure 2. Validation of the internal resistance fitting accuracy: (**a**) comparison between experimental and fitted resistance values and (**b**) residual distribution under different SOC and temperature conditions.

3. Physics-Informed Architecture and Algorithm Design

The standard PINN framework integrates physical laws by adding penalty terms (PDE, initial, and boundary losses) to the objective function. However, this soft constraint approach requires labor-intensive tuning of weight coefficients and often fails to strictly satisfy boundary conditions, particularly the Robin-type heat exchange at the battery surface. To eliminate boundary leakage and improve convergence, this study proposes an HCPINN architecture that enforces physical constraints through solution space reconstruction rather than simple loss-based penalties.

3.1. Construction of Solution Structure via Distance Functions

To ensure the predicted temperature field $\hat{T}(x,t)$ strictly adheres to the Robin boundary conditions defined in (3) without relying on penalty terms, a trial solution structure is constructed. Based on the theory of distance functions, the neural network output $\mathcal{N}(x,t;\theta)$ is transformed into a constrained space. The general form of the reconfigured solution is expressed as:

$$\hat{T}(x,t) = \mathcal{B}(x,t) + \mathcal{G}(x)\cdot\mathcal{N}(x,t;\theta) \tag{7}$$

where $\mathcal{B}(x,t)$ is the boundary augmentation term that captures the non-homogeneous boundary information, and $\mathcal{G}(x)$ is the spatial smoothing operator which ensures that the influence of the unconstrained network $\mathcal{N}$ is modulated near the boundaries.

In order to transform the aforementioned non-homogeneous conditions into a homogeneous form, let the augmentation term be a spatially constant ambient reference temperature:

$$\mathcal{B}(x,t) = T_{amb} \tag{8}$$

This selection converts the original Robin boundary conditions (3) into a homogeneous formulation concerning the temperature rise component $\Delta T = \hat{T}(x,t) - T_{amb}$, thereby requiring that the term $\mathcal{G}(x)\cdot\mathcal{N}(x,t;\theta)$ must automatically match the heat dissipation gradient at the boundaries.

Incorporate the proportional relationship between the coefficients h_0 and h_L at the battery terminals, the thermal conductivity λ directly into the operator $\mathcal{G}(x)$:

$$\mathcal{G}(x) = [1 + \frac{h_0}{\lambda}x + \frac{h_L}{\lambda}(L-x)]\cdot\mathcal{D}(x) \tag{9}$$

where $\mathcal{D}(x)$ is a reference distance function that equals zero at $x = 0$ and $x = L$, and here it is taken as $\mathcal{D}(x) = x^2(L-x)^2$.

Intuitively, $\mathcal{G}(x)$ acts as a boundary modulation function. The distance term $\mathcal{D}(x) = x^2(L-x)^2$ becomes zero at both axial boundaries, so it can suppress or reshape the free neural-network output near $x = 0$ and $x = L$, while preserving flexibility inside the battery. By embedding the ratio between the convection coefficient h and the axial thermal conductivity k_z, $\mathcal{G}(x)$ ensures that the reconstructed temperature field follows the Robin heat-exchange relation at the boundaries. Thus, $\mathcal{G}(x)$ can be viewed as an analytical correction layer that maps the raw neural-network output to a physically admissible solution.

The boundary consistency can be verified by differentiating the reconstructed solution at $x = 0$ and $x = L$, as shown in (10) and (11).

The rigor of the above design lies in the fact that by utilizing automatic differentiation to differentiate the reconstructed solution structure at boundary points, the following mapping logic can be obtained:

$$\left.\frac{\partial\hat{T}}{\partial x}\right|_{x=0} = \left.\frac{\partial\mathcal{B}(0,t)}{\partial x}\right|_{x=0} + \left[\mathcal{G}'(0)\cdot\mathcal{N}(0,t;\theta) + \mathcal{G}(0)\left.\frac{\partial\mathcal{N}(0,t;\theta)}{\partial x}\right|_{x=0}\right] \tag{10}$$

$$\left.\frac{\partial\hat{T}}{\partial x}\right|_{x=L} = \left.\frac{\partial\mathcal{B}(L,t)}{\partial x}\right|_{x=L} + \left[\mathcal{G}'(L)\cdot\mathcal{N}(L,t;\theta) + \mathcal{G}(L)\left.\frac{\partial\mathcal{N}(L,t;\theta)}{\partial x}\right|_{x=L}\right] \tag{11}$$

Since $\mathcal{G}(0) = \mathcal{G}(L) = 0$, the predicted gradient at the boundary is entirely determined by the product of the analytical derivative of the operator itself $\mathcal{G}'$, and the output of the neural network. By implementing a structured design for $\mathcal{G}'$, it can be ensured that the normal partial derivative of $\hat{T}$ is strictly locked to the slope determined by the physical

parameters $\frac{h}{\lambda}$ at any training stage, independent of the weight parameters of the neural network $\mathcal{N}(x,t;\theta)$. This reconstruction scheme based on homogenized mapping achieves a leap from fitting physical quantities to capturing the physical structure.

Specifically, to satisfy the heat transfer coefficient h and axial thermal conductivity λ, the reconstructed output is finally mapped as:

$$\hat{T}(x,t) = \text{Trans}[\mathcal{N}(x,t;\theta), \mathcal{B}(x), \mathcal{G}(x,t)] \tag{12}$$

Through this architecture, the network's solution space is restricted to a manifold where the prescribed mathematical boundary conditions are satisfied by construction. Consequently, the boundary loss becomes zero by construction, allowing the optimizer to focus entirely on the PDE residuals and initial states, thereby significantly accelerating convergence and enhancing estimation precision near the battery terminals.

This implementation distinguishes the proposed HCPINN from most existing PINN-based battery thermal models. In standard PINNs, the Robin boundary condition is typically included as a residual term in the loss function, and the boundary accuracy depends on the relative weight assigned to this term. In contrast, the proposed method modifies the trial solution itself. The neural network no longer directly outputs an arbitrary temperature field; instead, its output is analytically transformed into a temperature field that already satisfies the Robin heat-exchange relation at the two axial terminals. Therefore, the boundary condition is not learned approximately during optimization, but is imposed as a structural identity. This is the key methodological difference between the proposed HCPINN and conventional soft-constrained PINNs.

It should be noted that the hard-constraint formulation guarantees exact satisfaction of the prescribed Robin boundary condition within the mathematical model, rather than absolute physical consistency under all practical conditions. In real battery experiments, uncertainties in the heat transfer coefficient, axial thermal conductivity, sensor calibration, thermal contact resistance, and simplified one-dimensional modeling assumptions may still affect the physical fidelity of the reconstructed boundary behavior. Therefore, the term "physical consistency" in this work refers to consistency with the specified governing equations and boundary conditions used in the model.

3.2. Construction of Loss Function Under Hard Constraint Mechanism

In the proposed HCPINN framework, while the Robin boundary conditions are strictly satisfied via the architectural reconfiguration, the network must still be supervised by experimental measurement to ensure real-world fidelity. Therefore, the total loss function L_{total} is composed of the PDE residual loss, the initial condition loss, and a data-driven loss term:

$$L_{total} = \omega_{pde} L_{pde} + \omega_{ic} L_{ic} + \omega_{data} L_{data} \tag{13}$$

where ω_{pde}, ω_{ic}, and ω_{data} represent the weighting coefficients for each respective objective. To quantify these loss components, the following sets of points are defined within the spatiotemporal domain.

The PDE loss L_{pde} ensures the reconstructed temperature field $\hat{T}(x,t)$ adheres to the energy conservation law at N_{pde} collocation points sampled from the interior domain. N_{pde} represents the number of points used to enforce the physical laws throughout the axial space and time:

$$L_{pde} = \frac{1}{N_{pde}} \sum_{i=1}^{N_{pde}} \left| \rho C_p \frac{\partial \hat{T}(x,t)}{\partial t} - \lambda \frac{\partial^2 \hat{T}(x,t)}{\partial x^2} - \dot{Q}(t) \right|^2 \tag{14}$$

where the partial derivatives are computed via Automatic Differentiation (AD).

To anchor the thermal evolution at the starting time ($t = 0$), define N_{ic} as the number of spatial points sampled across the axial length of the battery at the initial timestamp:

$$L_{ic} = \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} \left| \hat{T}(x_j, 0) - T_{amb} \right|^2 \tag{15}$$

In real-world scenarios, sensor data is often limited. The total number of experimental measurement points N_{data} is introduced, representing the sparse observations available from physical sensors (e.g., surface thermocouples):

$$L_{data} = \frac{1}{N_{data}} \sum_{i=1}^{N_{data}} \left| \hat{T}(x_d, t_d) - T_{exp}(x_d, t_d) \right|^2 \tag{16}$$

where (x_d, t_d) are the specific coordinates where sensors are located. By eliminating the boundary loss through hard constraints, the optimizer can more effectively balance the trade-off between the physical PDE (N_{pde} points) and the experimental data (N_{data} points), leading to a high-fidelity reconstruction of the internal temperature field even with highly sparse sensing.

3.3. Physics-Informed Neural Network Structure

The neural network architecture of the HCPINN is designed to map multi-source electro-thermal inputs to a physically consistent temperature field through a rigorous structural identity. The input layer of the network receives the spatial coordinate x, time t, and the real-time electrical parameters including current I and terminal voltage U. These four dimensions (x, t, I, U) are processed through a sequence of 4 to 6 fully connected hidden layers, each containing 20 to 50 neurons, to capture the complex nonlinearities of the battery's thermal dynamics. Within these hidden layers, the hyperbolic tangent activation function is employed to ensure infinite-order differentiability, which is a prerequisite for computing the high-order partial derivatives required by the automatic differentiation process. Crucially, the direct output of the neural network, defined as $\mathcal{N}(x, t; \theta)$, does not represent the physical temperature. Instead, it serves as an original mapping quantity that is subsequently utilized to construct the physical solution.

The physical temperature field $\hat{T}$ is analytically reconstructed through the structural identity defined in (12), which integrates the operator $\mathcal{G}(x)$ and the boundary augmentation term $\mathcal{B}(x, t)$. This mathematical construction ensures that the boundary conditions are satisfied by identity, meaning that the predicted temperature field inherently adheres to the boundary laws regardless of the network parameters θ. Consequently, the traditional boundary loss term is entirely removed from the optimization objective. The training process focuses exclusively on minimizing the **PDE residual loss** L_{pde}, the **data-driven loss** L_{data} at sparse measurement points (x_d, t_d), and the initial condition loss L_{ic}. As illustrated in the topological diagram of the HCPINN structure in Figure 3, the spatiotemporal and electrical inputs are fused within this constrained manifold to ensure the global consistency of the predicted thermal distribution.

To further enhance the fidelity of the internal thermal representation, the HCPINN embeds the measured current I, internal resistance R, and voltage U directly into the PDE residual loss to satisfy the nonlinear heat generation mechanism defined in (4) and (5). By incorporating these dynamic electrical sequences, the physical constraint term provides a precise description of the heat source intensity under varying discharge rates. This embedding strategy ensures that the gradient updates of the network are guided by the coupled electro-thermal physics, allowing the model to bridge the gap between the original

mapping quantity $\mathcal{N}(x,t;\theta)$ and the real-world thermal process. Through this structural approach, the HCPINN achieves high-fidelity reconstruction of the axial temperature field using only sparse surface sensor data, while maintaining strict adherence to the underlying energy conservation laws.

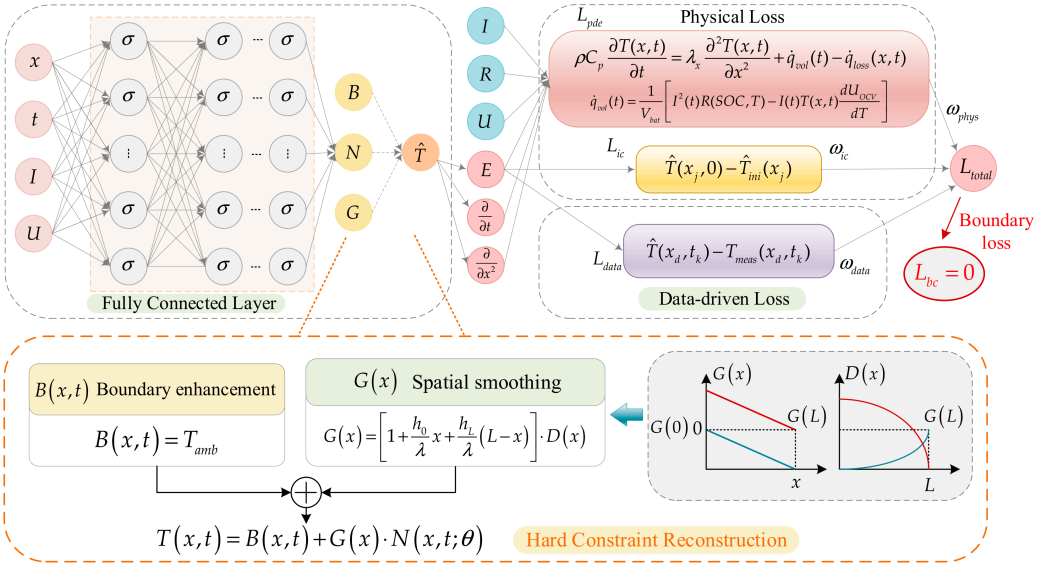


Figure 3. Structural topology of the HCPINN.

4. Experimental Validation and Analysis

To validate the reconstruction capability of the HCPINN model for real battery thermal behavior, an experimental platform for axial temperature acquisition of an 18,650 lithium-ion battery was established, as shown in Figure 4. In the experiment, the battery was placed in a thermal chamber to maintain a stable ambient temperature. NTC temperature sensors were arranged along the axial direction, and the temperature variations at different positions were recorded in real time by a temperature acquisition device.

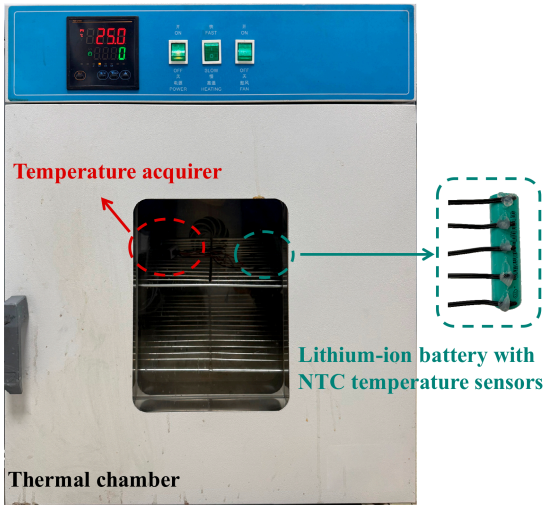


Figure 4. Experimental platform for axial temperature acquisition of the 18,650 lithium-ion battery.

This platform provides sparse axial temperature observations for the training and validation of the HCPINN model. Based on the collected data, the following analysis evaluates the temperature field reconstruction accuracy, error distribution, and axial thermal gradient characteristics of the model under different discharge rates.

4.1. Model Training and Experimental Configuration

The experimental validation of the HCPINN framework is performed using a Deliow 18,650 lithium-ion battery, the physical characteristic parameters of which are detailed in Table 1. During the testing process, the battery is housed within a thermal chamber maintained at a constant temperature of 25 °C to ensure a stable ambient environment. The experimental protocols involve complete discharge cycles conducted at three distinct rates: 1C, 2C, and 4C. To capture the precise axial temperature distribution, five NTC thermocouples are strategically positioned along the height of the battery at 0, 0.25 L, 0.5 L, 0.75 L, and L, providing high-fidelity data for both model training and performance validation. It should be noted that this five-sensor configuration is mainly used to provide sufficient reference measurements for validating the reconstructed axial temperature field, rather than representing the minimum sensing requirement for practical BMS deployment.

Table 1. Table of physical parameters for the experimental battery.

Parameter Name	Symbol	Unit	Value
Battery Mass	m	kg	0.045
Battery Density	ρ	$\mathrm{kg/m^3}$	2770
Rated Capacity	Q	Ah	2.5
Heat Capacity	C_p	$\mathrm{J/(kg{\cdot}K)}$	1020
Axial Thermal Conductivity	λ	$\mathrm{W/(m{\cdot}K)}$	15.7
Heat Transfer Coefficient	h	$\mathrm{W/(m^2{\cdot}K)}$	5.5

Note: The rated capacity was obtained from the manufacturer datasheet. The thermal parameters were initialized from literature values for commercial 18,650 cells, while k_z and h were further calibrated using the 1C temperature response and natural cooling curve, respectively.

To improve the repeatability and reliability of the experimental measurements, additional calibration and uncertainty control procedures were adopted. Before the discharge experiments, all NTC thermocouples were calibrated in a thermostatic chamber at 25 °C, 35 °C, and 45 °C using a reference thermometer with an accuracy of ±0.05 °C. The maximum calibration deviation of the temperature sensors was within ±0.15 °C, and this value was considered as the sensor calibration uncertainty in the subsequent analysis. To reduce the influence of thermal contact resistance, each thermocouple was tightly attached to the cell surface using thermally conductive silicone grease and high-temperature insulation tape. After installation, the sensors were kept in contact with the cell surface for at least 10 min before testing to ensure stable thermal coupling.

During all tests, the temperature and electrical signals were recorded at a sampling frequency of 1 Hz. Each discharge condition, including 1C, 2C, and 4C, was repeated three times under the same experimental configuration. The final data used for model training and validation were obtained by averaging the repeated measurements, while the standard deviation among repeated tests was used to evaluate experimental repeatability. The ambient temperature of the thermal chamber was controlled at 25 °C with a stability of ±0.3 °C throughout the experiments. These procedures ensured that the measured axial temperature data were sufficiently reliable for training and validating the proposed HCPINN model.

The scale and selection of the dataset were further specified as follows. Since the temperature and electrical signals were sampled at 1 Hz, the complete 1C, 2C, and 4C discharge processes contained 3601, 1801, and 901 synchronized time records, respectively, including the initial state. With five axial thermocouples, the averaged dataset used for modeling therefore contained 6303 synchronized electrical-state records and 31,515 temperature observations. Considering the three repeated tests, the raw experimental dataset before averaging contained 18,909 synchronized electrical-state records and 94,575 temperature observations. For each discharge rate, the averaged time records were divided into

training, validation, and test sets at a ratio of 70%:15%:15%. To avoid a biased evaluation caused by using only one continuous SOC interval as the test set, the split was performed by uniformly sampling along the full discharge trajectory, so that high-, medium-, and low-SOC states were all represented in each subset. Specifically, the numbers of time records used for training/validation/testing were 2521/540/540 for 1C, 1261/270/270 for 2C, and 631/135/135 for 4C, respectively. The same indices were applied to the five temperature sensors and the synchronized current-voltage-SOC inputs. Only the training subset was used in the supervised temperature loss, the validation subset was used for monitoring convergence and hyperparameter selection, and the held-out test subset was used only for the final error evaluation reported in this study.

To clarify the specific state of the battery data used in this study, the electrical and thermal measurements were synchronized according to the discharge SOC. The rated capacity of the tested cell was 2.5 Ah; therefore, the 1C, 2C, and 4C discharge currents corresponded to 2.5 A, 5.0 A, and 10.0 A, respectively. Each test started from a fully charged state (SOC = 100%) after thermal equilibration at 25 °C and ended at the discharge cut-off state. The SOC used for model input and data alignment was calculated by coulomb counting, with SOC $= 100\% - (Q_d/C^n) \times 100\%$, where Q_d is the discharged capacity and C^n is the rated capacity.

Figure 5 shows the representative voltage-SOC/capacity curves during discharge and the corresponding SOC-resolved temperature samples. The marked SOC states (100%, 80%, 60%, 40%, 20%, and 0%) indicate the electrical states at which the five axial temperature measurements were extracted for thermal-state comparison and model validation. As the SOC decreases, the terminal voltage exhibits the typical plateau followed by an end-of-discharge drop, while the temperature rise becomes more pronounced at higher discharge rates.

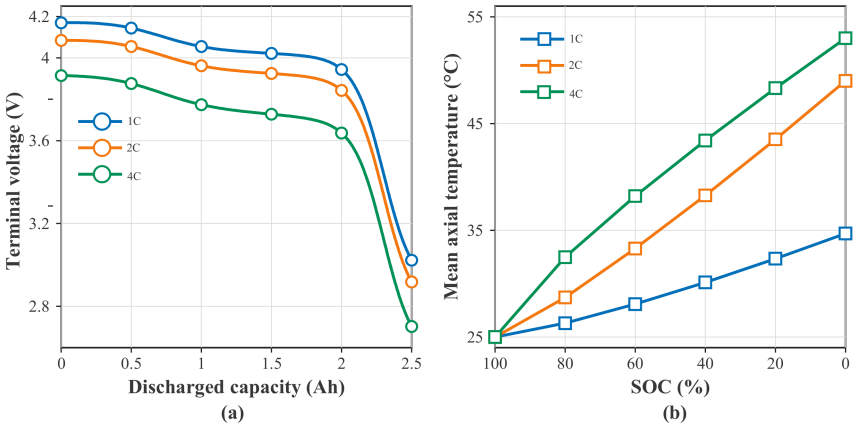


Figure 5. Voltage-SOC/capacity curves and SOC-resolved temperature sampling during discharge: (**a**) terminal voltage as a function of discharged capacity and SOC under 1C, 2C, and 4C rates; (**b**) mean axial temperature extracted from the five thermocouples at representative SOC states.

In terms of algorithmic implementation, the HCPINN is developed using the PyTorch 1.8.2 deep learning framework within a Python 3.8 environment. The computational tasks are executed on a high-performance workstation equipped with an Intel Core i9 processor and an NVIDIA GeForce RTX 3080 GPU. The network optimization follows a rigorous two-stage combined strategy involving the Adam and L-BFGS algorithms. Initially, the Adam optimizer is utilized for 10,000 epochs of coarse training, leveraging its stochastic gradient descent properties to effectively bypass local minima within the complex loss landscape. Following this initial phase, the optimization switches to the L-BFGS algorithm for second-order refinement. This stage continues until the physical residual L_{pde} reaches a

predefined convergence threshold, ensuring that the final reconstructed temperature field strictly adheres to the underlying thermal physics while achieving high numerical precision.

4.2. Axial Temperature Rise Evolution

Figure 6 illustrates the axial global temperature evolution process of the battery during discharge at a 1C rate, starting from an initial room temperature of 25 °C and ending at the 3600s cutoff time. By comparing the experimental measured data (Figure 6a) with the PINN-estimated data (Figure 6b), it can be observed that both exhibit a high degree of consistency in their spatiotemporal distribution characteristics.

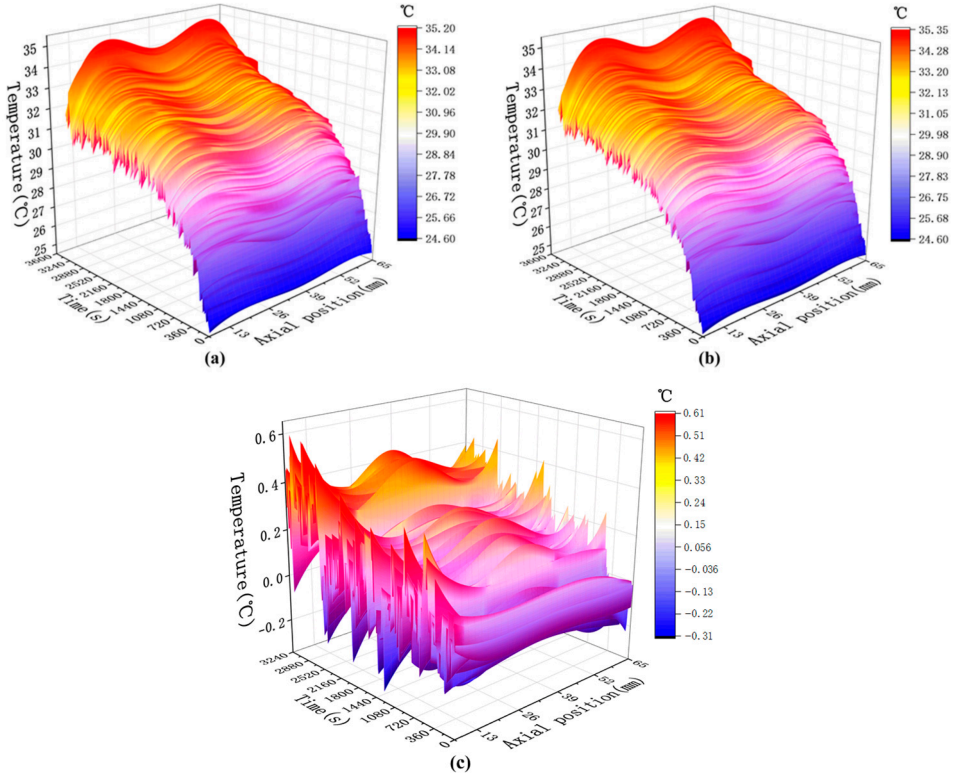


Figure 6. Temperature distribution evolution diagram at 1C rate: (**a**) measured data, (**b**) HCPINN-estimated data and (**c**) estimation error.

The maximum temperature across the entire axial domain of the battery steadily rises from 25 °C to approximately 35 °C, with an overall temperature increase of about 10 °C. The network reconstructs this smooth temperature rise surface, covering the axial length from x = 0 (positive electrode) to x = 65 mm (negative electrode). This indicates that, under low heat generation intensity, the model can accurately extract the characteristics of ohmic heat and reaction heat as described in (4), combined with the hard-constrained boundary conditions, generates a temperature response field with high physical consistency.

As the discharge process enters its final stage (after approximately 3000s), the internal resistance exhibits nonlinear growth due to the decrease in the battery's state of charge (SOC), causing slight fluctuations in the experimentally measured temperature rise slope. At this point, although the estimation error of the network increases somewhat, the maximum axial deviation is only +0.6 °C.

Figure 7 presents the reconstructed results of the axial temperature field of the battery under a 2C discharge rate. Compared to the 1C operating condition, the 2C discharge exhibits more intense kinetic characteristics.

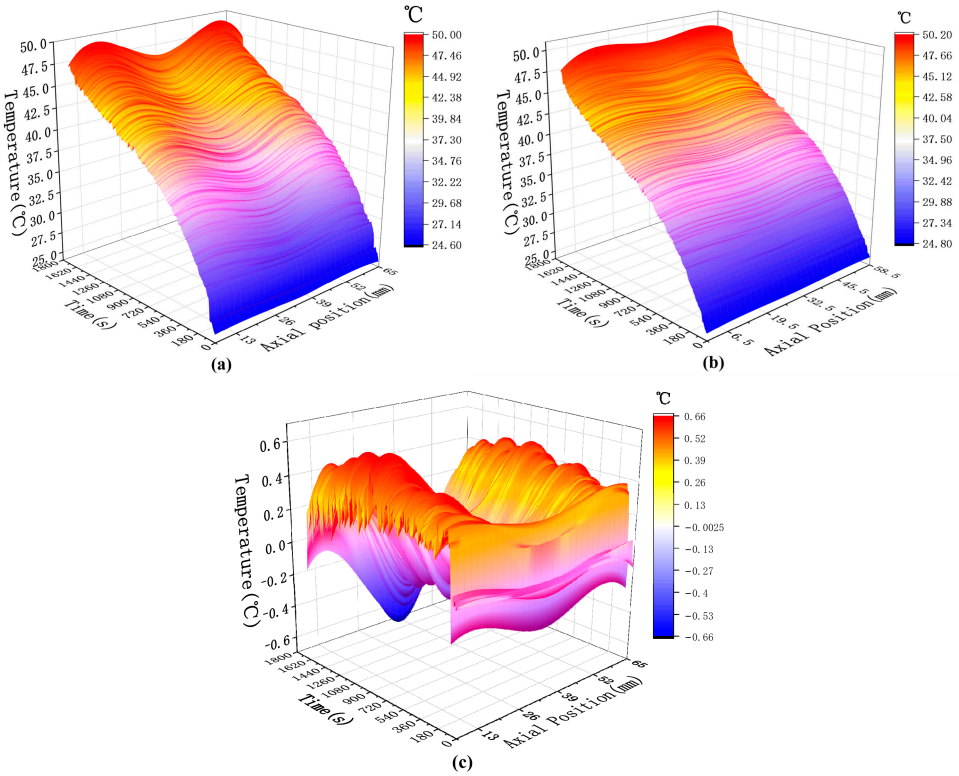


Figure 7. Temperature distribution evolution diagram at 2C rate: (**a**) measured data, (**b**) HCPINN-estimated data and (**c**) estimation error.

Under 2C rate, the maximum temperature rise across the entire axial length of the battery climbs from the initial 25 °C to 49.6 °C at the end of discharge, with a net temperature increase of nearly 25 °C. This significant temperature rise not only tests the network's ability to map the source term $\dot{Q}(t)$ in the energy balance equation but also imposes higher demands on the physical sensitivity of the boundary heat transfer coefficient h_0 and h_L.

In terms of local error distribution, the 2C operating condition exhibits dynamic characteristics distinct from those at lower discharge rates. During the initial discharge phase (around 15 s), a noticeable increase in estimation error occurs, rapidly jumping from an initial value of approximately 0.1 °C to −0.3 °C. This initial error jump has a clear physical basis: At the instant when the 2C current is triggered, intense Joule heat shock and electrochemical polarization heat are generated inside the battery, causing a sharp increase in the local temperature rise slope. The network requires several iterative steps in the spatiotemporal coordinate mapping to adapt to this sudden thermal shock, resulting in a temporary lagging negative deviation.

As the discharge process stabilizes, the error enters a phase of steady evolution. By the end of discharge, the maximum axial temperature difference stabilizes at around −0.6 °C, with this deviation primarily concentrated at the midpoint (0.5 L) along the battery's axis. Analysis suggests that, since the central region of the battery is at the deepest point of axial heat conduction and experiences the strongest heat accumulation effect, the model's slight underestimation of the internal heat generation term in this region leads to the final deviation.

When the discharge rate further increases to 4C, the battery exhibits more pronounced nonlinear thermal behavior due to intensified Joule heating and polarization effects. Figure 8 illustrates the spatiotemporal evolution characteristics of the axial temperature field of the battery under this operating condition.

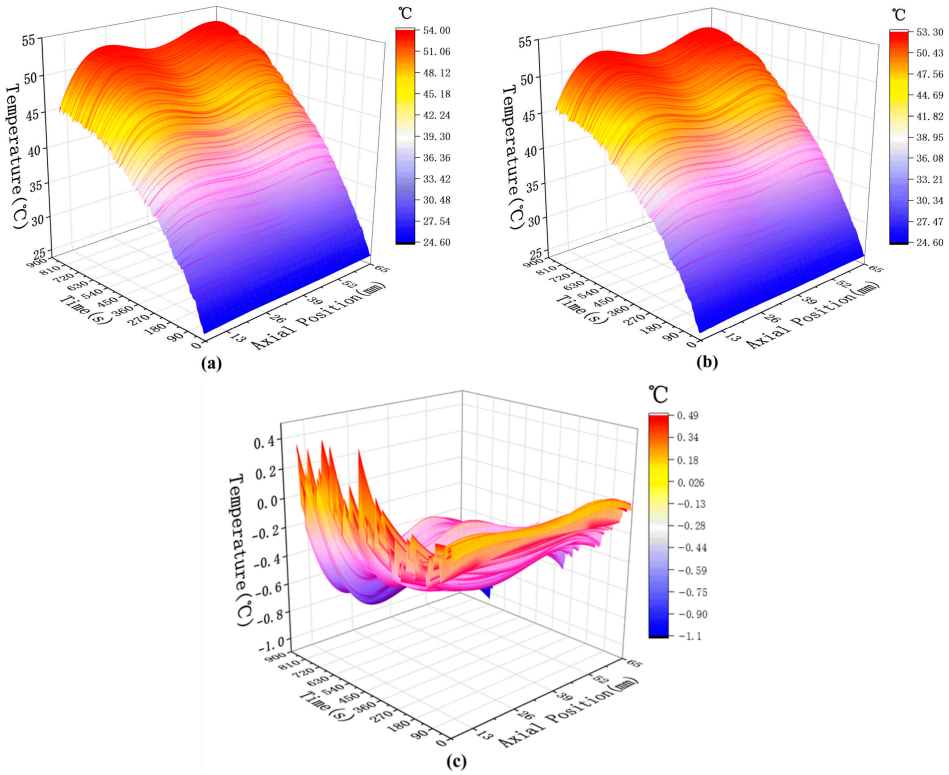


Figure 8. Temperature distribution evolution diagram at 4C rate: (**a**) measured data, (**b**) HCPINN-estimated data and (**c**) estimation error.

From Figure 8, a 4C discharge rate triggers substantial Joule heating and intense polarization responses. The maximum axial temperature of the battery soars from an initial 25 °C to approximately 53 °C in a very short time at the end of discharge (with a net temperature rise of 28 °C).

Significant axial thermal gradients are generated within the battery, with a notably widened temperature difference between the central region and the boundaries. In terms of error quantification, high-rate operating conditions pose a significant challenge to the sensitivity of the model's physical parameters. Experimental results indicate that the maximum estimation error at the end of 4C discharge occurs at x = 65 mm (at the negative current collector), with a deviation of −1.1 °C.

The maximum absolute error (Emax), mean absolute error (MAE), and root mean square error (RMSE) are introduced as quantitative evaluation metrics to assess the accuracy of the model in reconstructing the full-field temperature of lithium-ion batteries, with specific data presented in Table 2.

Table 2. Axial temperature estimation errors under different discharge rates.

Discharging Rate	Emax (°C)	MAE (°C)	RMSE (°C)
1C	0.6042	0.1327 ± 0.0184	0.1601 ± 0.0217
2C	0.6573	0.2540 ± 0.0268	0.2933 ± 0.0305
4C	1.0545	0.3375 ± 0.0342	0.3843 ± 0.0396

To evaluate the statistical robustness of the proposed HCPINN, each discharge experiment was repeated three times under the same experimental configuration, and the mean and standard deviation of MAE and RMSE were calculated. As shown in Table 2, the standard deviations remain relatively small under all discharge rates, indicating good repeatability of the experimental results. Although the error fluctuation slightly increases

at 4C due to stronger heat generation and more pronounced nonlinear thermal dynamics, the MAE remains below 0.34 °C, demonstrating that the proposed method maintains stable estimation performance across repeated tests.

To further demonstrate the superiority of the proposed HCPINN over conventional physics-based, data-driven, and filtering-based approaches, a comparative evaluation was conducted using the same experimental dataset and hardware platform. The baseline models include a standard soft-constrained PINN, ANN, LSTM, an equivalent circuit thermal model, and a Kalman filter-based estimator. The comparison results are summarized in Table 3.

Table 3. Comparison of the proposed HCPINN with baseline temperature estimation models.

Method	MAE (°C)	RMSE (°C)	Training Time
Equivalent circuit thermal model	0.6812	0.8125	-
Kalman filter-based method	0.5746	0.6953	-
ANN	0.5368	0.6427	18.6 min
LSTM	0.4625	0.5489	31.4 min
Standard PINN	0.3579	0.4268	64.7 min
Proposed HCPINN	0.2414	0.2792	51.3 min

To further clarify the experimental procedure of the control group, the standard PINN was selected as the primary control model because it has the same network inputs, training data, and physical governing equation as the proposed HCPINN, while the Robin boundary condition is imposed through a soft penalty term in the loss function. In contrast, the proposed HCPINN embeds the Robin boundary condition into the trial solution structure, thereby removing the boundary loss from the optimization objective. As shown in Figure 9a, the proposed HCPINN exhibits a faster and more stable convergence trend than the standard PINN. Figure 9b further presents the evolution of different loss components, where the boundary loss remains nearly zero due to the hard-constraint formulation. The MAE and RMSE comparisons in Figure 10 demonstrate that the proposed HCPINN achieves lower estimation errors than the standard PINN under all discharge rates, confirming the effectiveness of the hard-constraint strategy.

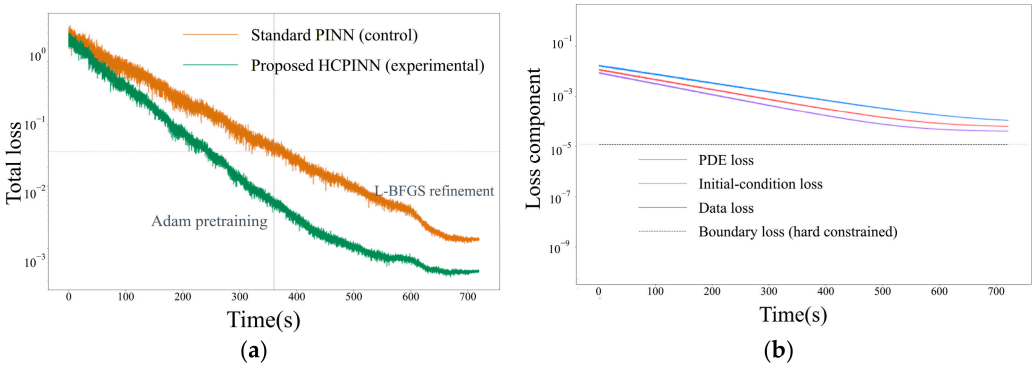


Figure 9. Training process comparison between the standard PINN baseline and the proposed HCPINN: (**a**) total loss convergence curves and (**b**) loss component evolution of the proposed HCPINN.

As shown in Table 3, the proposed HCPINN achieves the lowest MAE and RMSE among all compared methods. The equivalent circuit thermal model and Kalman filter-based estimator exhibit relatively high computational efficiency, but their simplified thermal representations limit their capability to capture the axial temperature gradient inside the cylindrical cell. Pure data-driven models, including ANN and LSTM, show improved fitting capability, while their lack of physical constraints leads to reduced robustness under

high-rate discharge conditions. Compared with the standard PINN, the proposed HCPINN further reduces the average MAE and RMSE by embedding the Robin boundary conditions into the network architecture rather than imposing them through soft penalty terms. This hard-constraint design eliminates boundary leakage, improves convergence efficiency, and enhances the physical consistency of the reconstructed temperature field.

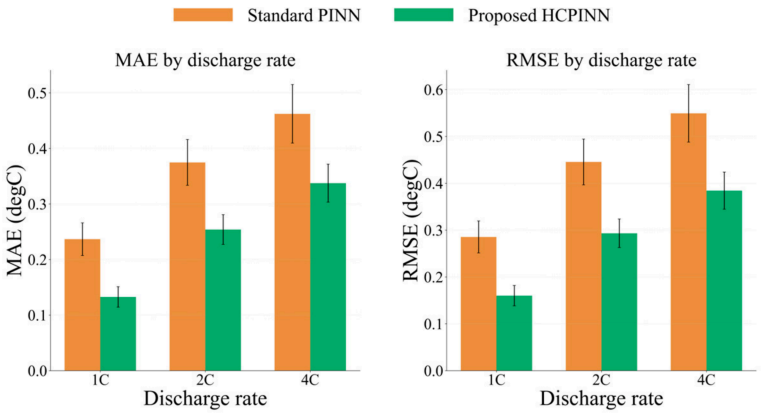


Figure 10. MAE and RMSE comparisons under different discharge rates.

4.3. Sparse-Sensor Reconstruction Analysis

Since commercial BMSs often use only one or two surface temperature sensors due to cost, packaging space, and wiring constraints, the five-sensor configuration in this study may still be relatively dense for practical applications. Therefore, to further evaluate the applicability of the proposed HCPINN under more realistic sparse-sensor conditions, an additional reduced-sensor experiment was conducted. In this analysis, the original five-sensor configuration was treated as the reference setting, while two reduced configurations were constructed by selectively using fewer sensor observations for model training. Specifically, the three-sensor case used the measurements at x = 0, 0.5L, and L, while the two-terminal case only used the measurements at x = 0 and x = L. The remaining internal sensor locations were reserved for validation to assess the reconstruction accuracy of the central temperature field.

The comparison results are summarized in Table 4. It can be observed that reducing the number of sensors inevitably increases the estimation error, especially near the central region where direct measurements are unavailable. Nevertheless, even under the two-terminal sensing condition, the HCPINN still maintains acceptable reconstruction accuracy due to the embedded heat conduction equation, electro-thermal heat generation model, and hard-constrained Robin boundary conditions. This indicates that the physical constraints help compensate for the lack of internal measurements and enable the model to infer the overall axial temperature evolution from highly sparse boundary observations.

Table 4. Reconstruction accuracy under different sensor configurations.

Sensor Configuration	Training Sensor Locations	MAE (°C)	RMSE (°C)	Central Hot-Spot Error (°C)
Five-sensor setting	0, 0.25L, 0.5L, 0.75L, L	0.2414	0.2792	0.31
Three-sensor setting	0, 0.5L, L	0.2865	0.3412	0.38
Two-terminal setting	0, L	0.3928	0.4687	0.56

Compared with the five-sensor setting, the three-sensor configuration shows only a moderate increase in MAE and RMSE, indicating that retaining the central sensor provides useful information for capturing the dominant hot-spot evolution. When only the two terminal sensors are used, the MAE and RMSE increase to 0.3928 °C and 0.4687 °C,

respectively. The central hot-spot error also increases to 0.56 °C because the model no longer has direct access to the central temperature measurement during training. However, this error remains within a reasonable range for real-time thermal monitoring, especially considering that the model relies only on boundary temperature observations.

The results suggest that the HCPINN can reconstruct the central thermal behavior under highly sparse sensing conditions, but the accuracy is affected by the observability of internal temperature gradients. The hard-constrained boundary formulation ensures physically consistent heat exchange at both terminals, while the PDE residual and electro-thermal heat source model provide additional constraints for inferring internal temperature evolution. Therefore, even with two terminal sensors, the model can capture the overall axial temperature trend and estimate the central hot-spot region with acceptable accuracy. For commercial BMS deployment, the two-terminal configuration provides a practical compromise between sensor cost and reconstruction performance, while a three-sensor configuration may be preferred when higher hot-spot accuracy is required.

4.4. Computational Cost and Real-Time Feasibility Analysis

For real-time battery management system (BMS) applications, computational efficiency is as important as estimation accuracy. Therefore, the computational cost of the proposed HCPINN was further evaluated in terms of training time, prediction time, hardware requirements, and embedded deployment feasibility. All experiments were conducted on a workstation equipped with an Intel Core i9 CPU and an NVIDIA GeForce RTX 3080 GPU, using the PyTorch framework in a Python 3.8 environment.

The computational cost of the proposed model is summarized in Table 5. The offline training process, including Adam-based pretraining and L-BFGS refinement, required approximately 51.3 min. Although the training stage is more time-consuming than pure data-driven ANN and LSTM models, it is performed offline and does not affect real-time BMS operation. Once trained, the HCPINN only requires a single forward propagation to estimate the temperature at a given spatiotemporal point. The average prediction time is approximately 0.34 ms per sample on GPU and 1.86 ms per sample on CPU, which satisfies the typical real-time sampling requirement of battery thermal monitoring.

Table 5. Computational cost and real-time feasibility of the proposed HCPINN.

Item	Result
Training hardware	Intel Core i9 CPU + NVIDIA RTX 3080 GPU
Software environment	Python 3.8 + PyTorch
Training strategy	Adam + L-BFGS
Offline training time	51.3 min
GPU prediction time	0.34 ms/sample
CPU prediction time	1.86 ms/sample
Model input	x, t, I, V
Model output	T(x,t)
Embedded feasibility	Feasible after model compression

The above results indicate that the main computational burden of HCPINN lies in the offline training stage, where automatic differentiation is required to calculate the PDE residuals. In contrast, the online inference stage does not involve PDE residual optimization or gradient calculation. Therefore, the proposed model can operate as a lightweight forward estimator after training. For embedded BMS applications, the trained network can be further compressed by network pruning, parameter quantization, or knowledge distillation. In addition, because the input dimension is low and the network consists only of fully connected layers, the model can be deployed on high-performance automotive

microcontrollers or edge computing units. These results demonstrate that the proposed HCPINN is computationally feasible for real-time battery temperature monitoring.

4.5. Physical Boundary Analysis of Cooling Strategies

Building upon the high-fidelity spatiotemporal reconstruction capabilities of the HCPINN, the continuous field information output by the network is utilized to further invert the ideal cooling requirements of the battery during actual operation, thereby providing a quantitative design criterion for BTMS.

By integrating the boundary temperature gradient $\frac{\partial \hat{T}}{\partial x}$ captured by the network with the axial thermal conductivity λ, the critical convection heat transfer coefficient $\hat{h}$ required to maintain thermal equilibrium can be inverted based on the Robin boundary condition (3), as illustrated in (17).

$$\hat{h} = \frac{-\lambda \nabla \hat{T} \cdot \vec{n}}{\hat{T}_{bound} - T_{amb}} \tag{17}$$

where $\nabla \hat{T} \cdot \vec{n}$ denotes the normal gradient at the boundary, and $\hat{T}_{bound}$ represents the real-time boundary temperature reconstructed by the model.

This inversion process, as illustrated through the dynamic evolution curves of $\hat{h}$ for 1C, 2C, and 4C discharge rates, allows for the transformation of abstract thermal gradients into actionable engineering parameters. The inversion curves are shown in Figure 11.

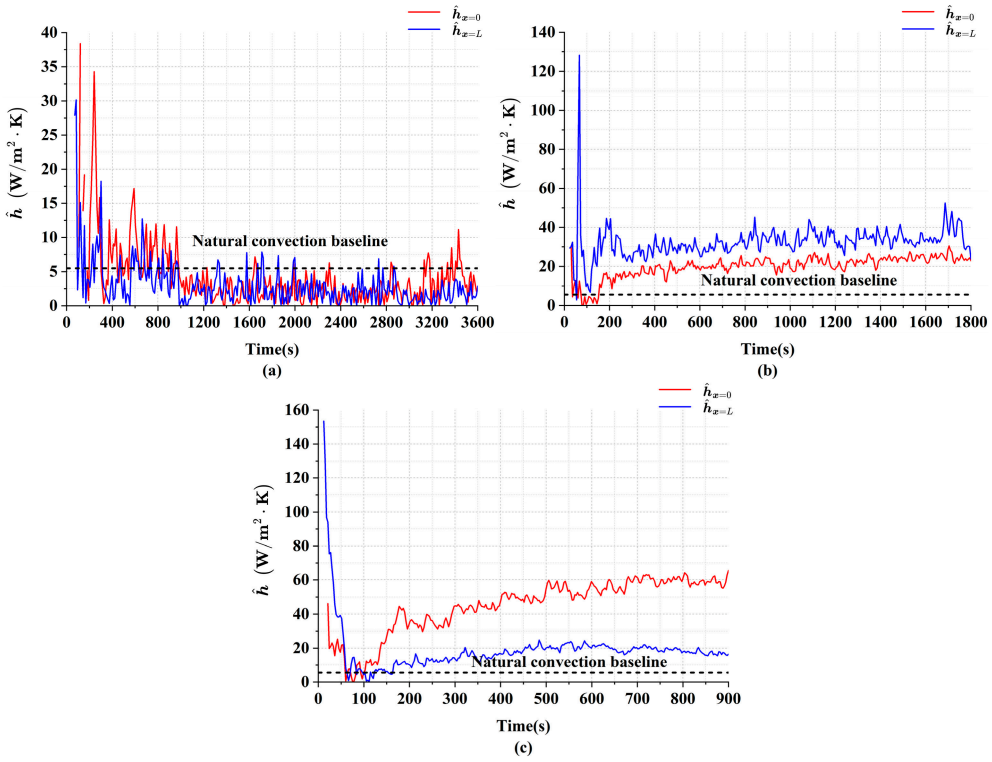


Figure 11. Critical convection heat transfer coefficient $\hat{h}$ under different rates: (**a**) 1C, (**b**) 2C and (**c**) 4C.

The analysis of the inversion curves reveals that the cooling demand exhibits significant non-linear growth characteristics as the discharge rate increases. For the 1C and 2C scenarios, the inverted $\hat{h}$ values remain relatively stable and reside within the typical range for natural convection or low-speed air cooling, indicating that the internal heat generation rate and axial conduction capacity maintain a sustainable dynamic balance at these lower intensities. However, under the 4C high-rate condition, the $\hat{h}$ curve climbs sharply as the depth of discharge (DOD) increases, with its average value far exceeding the physical heat transfer limits of conventional passive air cooling. This trend underscores

that under high-rate loads, passive cooling alone is insufficient to suppress the polarization of the spatial temperature gradient, necessitating the introduction of high-power active cooling strategies to ensure axial temperature uniformity. Ultimately, the spatiotemporal field reconstruction based on **HCPINN** enables comprehensive thermal state monitoring and converts physical gradients into direct indicators for cooling strategy design, laying a theoretical foundation for intelligent battery thermal management systems with high dynamic response and physical perception.

It should be noted that the inverted critical convection heat transfer coefficient h_{crit} was used in this study as a physics-informed indicator for evaluating the cooling demand, rather than as a directly validated closed-loop control parameter. The present experimental platform was designed for axial temperature acquisition under controlled natural-convection conditions, and no active cooling actuator was included to dynamically impose the inverted h_{crit}. Therefore, the ability of h_{crit} to guide a practical BTMS toward thermal equilibrium has not yet been experimentally verified in this work. Nevertheless, the inversion result provides a quantitative reference for determining the required cooling intensity under different discharge rates. Future work will combine CFD simulation and controllable air/liquid cooling experiments to validate whether the cooling strategy designed according to h_{crit} can effectively reduce axial temperature gradients and maintain thermal equilibrium under practical BTMS conditions.

5. Conclusions

This study developed a Hard-Constrained Physics-Informed Neural Network (HCPINN) for reconstructing the axial temperature field of lithium-ion batteries. By embedding the Robin boundary conditions into the network solution structure through a distance-function-based formulation, the proposed method enforces boundary consistency by construction rather than by soft penalty terms. This design reduces the convergence instability commonly observed in conventional PINNs and improves the consistency of the reconstructed temperature field with the prescribed thermal boundary model.

Experimental validation on Deliow 18,650 cells under different discharge rates shows that the HCPINN can accurately estimate continuous axial temperature distributions using sparse surface measurements. Compared with pure data-driven models and standard soft-constrained PINNs, the proposed method achieves lower reconstruction errors while maintaining efficient online inference. In addition, the differentiable temperature field enables the inversion of the critical convection coefficient, providing a physics-informed indicator for evaluating cooling demand under different operating conditions.

Several limitations remain. The current model is based on one-dimensional axial heat conduction and does not explicitly capture radial gradients, jelly-roll anisotropy, tab heating, or local hot spots in practical cells. The experimental validation is also limited to fresh 18,650 cylindrical cells, and its generalization to larger cylindrical cells, pouch cells, and aged batteries has not yet been fully demonstrated. Moreover, full-lifecycle parameter drift caused by battery degradation, such as changes in internal resistance and entropy coefficients, is not explicitly modeled. The inverted critical convection coefficient is currently used as a cooling demand indicator rather than a directly validated closed-loop BTMS control parameter.

Future work will extend the HCPINN framework toward 2D/3D thermal field reconstruction, cell-to-cell pack modeling, and module-level temperature estimation. Further studies will also incorporate SOH-dependent thermal parameters, validate the framework on 21700 -type, 4680 -type, pouch cells, and aged batteries, and conduct closed-loop BTMS experiments with controllable air or liquid cooling actuators to verify the practical effectiveness of h_{crit}-guided thermal management.

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Data Availability Statement: The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request. The data are not publicly available due to project-related restrictions.

Conflicts of Interest: Authors Lingqing Guo, Kangliang Zheng, Xiucheng Wu, Jinhong Wang, Xiaofeng Lai, and Peiyuan Deng were employed by the Huizhou Power Supply Bureau, Guangdong Power Grid Corporation. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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